

Human-Centered Personas at Scale: Using LLMs and RAG to Build Personas from Mixed-Methods Data

Wymer, C., Wolfe, D., & Choe, A. (2026, April 30). Human-Centered Personas at Scale: Building LLM-Based Personas Using Mixed Methods Data [Master Tutorial]. Society for Industrial and Organizational Psychology Annual Conference, New Orleans, LA, United States

Presenters



Chelsea (Wymer) Homann, M.S.

Independent I-O Practitioner

The bridge between AI and behavioral science is how we understand and shape what's changing in people and organizations.



Diana Wolfe, PhD

VP, AI Research and Strategy

AI reshapes human systems, and someone needs to build the measurement science for how.



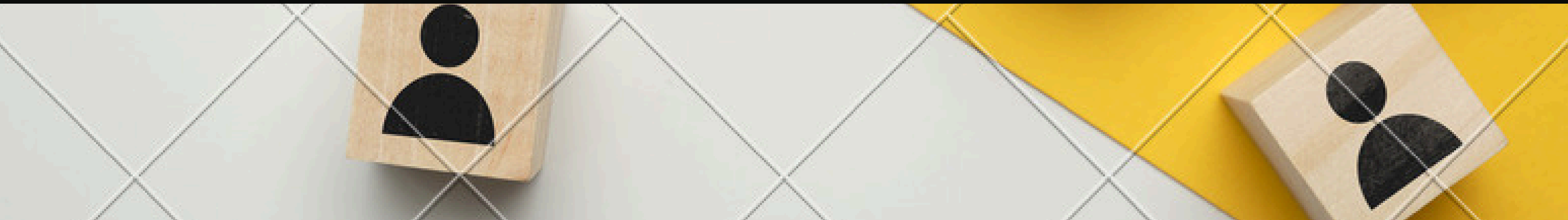
Alice Choe

Org Behavior, PhD Candidate

Before you lament how you're not seeing the impact you expected to see from AI, have you considered whether you're measuring the wrong thing?



A reproducible and practical way to turn
employee data into human-centered personas,
using AI (LLMs), rigorous statistical methods
(clustering), and real company context (RAG)



Original Article



We Reject the Use of Generative Artificial Intelligence for Reflexive Qualitative Research

Tanisha Jowsey ¹, Virginia Braun², Victoria Clarke³, Deborah Lupton ⁴, and Michelle Fine⁵

Abstract

Four hundred and nineteen experienced qualitative researchers from 32 countries invite readers of *Qualitative Inquiry* to consider their position on use of generative artificial intelligence (GenAI) for qualitative research. We hold the position that analytic approaches such as reflexive thematic analysis are human research practices requiring a subjective, positioned, and reflexive researcher and therefore the use of GenAI in such approaches is not methodologically congruent. We additionally reject GenAI for reflexive qualitative approaches on the grounds of social and environmental justice.

Qualitative researchers' AI rejection is based on identity, not reason

The claim that AI can't make meaning contradicts what researchers are finding when they put these tools to careful, critical use, says James Goh

Published on January 16, 2026

Last updated January 23, 2026

James Goh

The letter spread quickly, forwarded from colleague to colleague, often accompanied by notes of approval or unease. It was an open letter bearing 419 signatures from qualitative researchers across 38 countries. Its language left no room for negotiation: generative artificial intelligence is "inappropriate in all phases of reflexive qualitative analysis". Not sometimes. Not with safeguards. Always.



Source: Donald Iain Smith/Getty Images

That absolutism matters, because qualitative research is how institutions decide which human experiences count. Shut off tools at this level and you shape who gets heard in education, labour and international development.

Among the lead signatories of [the letter](#), published in the Sage journal *Qualitative Inquiry*, are Virginia Braun and Victoria Clarke, whose framework for reflexive thematic analysis has shaped how a generation of scholars thinks about interpretation. Their [foundational paper](#) has been cited nearly 300,000 times. When they speak, people listen. And what they are saying now is that the conversation about AI is over.

Learning Objectives



LO1

Turn real employee data into clear, research-based personas that reflect how people experience the organization

LO2

Use a clustering pipeline to generate personas grounded in actual evidence

LO3

Build a single dataset that combines quant and qual data so they can be analyzed jointly

LO4

Apply ethical frameworks to assess responsible use of AI and leadership applications

Cluster 0: The Anchored Advocates

Run ID: fresh_run_2026_04_26 | Timestamp: 2026-04-26

Statistical Fingerprint

Dimension	Z-score	Interpretation
Cared_About	+0.356	Moderate
Excited	+1.026	High
Helpful_Info	+0.170	Moderate
Trust_Leadership	+1.031	High
Morale	+0.741	High

Size: 3,426 respondents (34.3% of total) **Quality Grade:** B (Silhouette = 0.311) **Outlier rate:** 10.9% **Dominant demographic:** Business Unit: Product | Level: Senior | FLSA: Exempt | Tenure: 10+ yrs

LPA Psychological Fingerprint: High-Trust_Leadership / High-Excited / High-Morale

Personas

PERSONAS ARE RESEARCH-BASED PROFILES OF REPRESENTING GROUPS OF INDIVIDUALS. THEY TURN RESEARCH INTO SOMETHING TEAMS CAN UNDERSTAND AND ACT ON.

Three key functions

- Build empathy: turn data into human stories leaders can relate to
- Create alignment: create a shared understanding of employees across teams
- Accelerate change: help teams see the gap between intention and lived experience

How they're traditionally built:

- Deep qualitative analysis
- Iterative coding
- Expert validation



Challenges with Building Personas at Scale

Traditional approaches to building personas don't scale for 3 reasons.

Volume

Problem: **Too much data**
(thousands of documents → manual coding doesn't scale)

AI Potential: Code entire corpora in hours. Embedding-based clustering surfaces patterns that smaller sampling misses.

Variation

Problem: **Too much complexity**
(given different roles, regions, experiences, etc.)

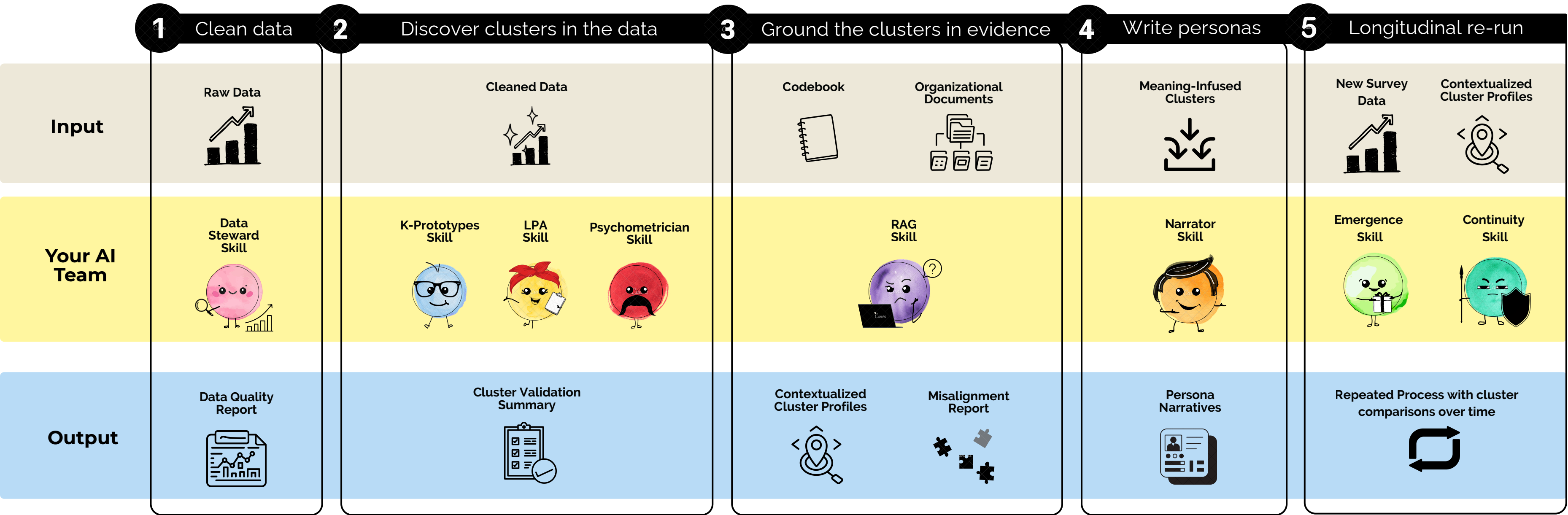
AI Potential: Composite personas preserve heterogeneity. Query across dimensions without pre-committing to archetypes.

Velocity

Problem: **Too slow**
(research cycles in months, behavior shifts in weeks)

AI Potential: Treat Personas as living instruments. Refresh cadence matches the rate of change.

End to End Process



You're still the scientist.

THIS IS A WORKFLOW WITH AUTOMATION, NOT AN AUTOMATED WORKFLOW. YOU'RE IN THE DRIVER'S SEAT.

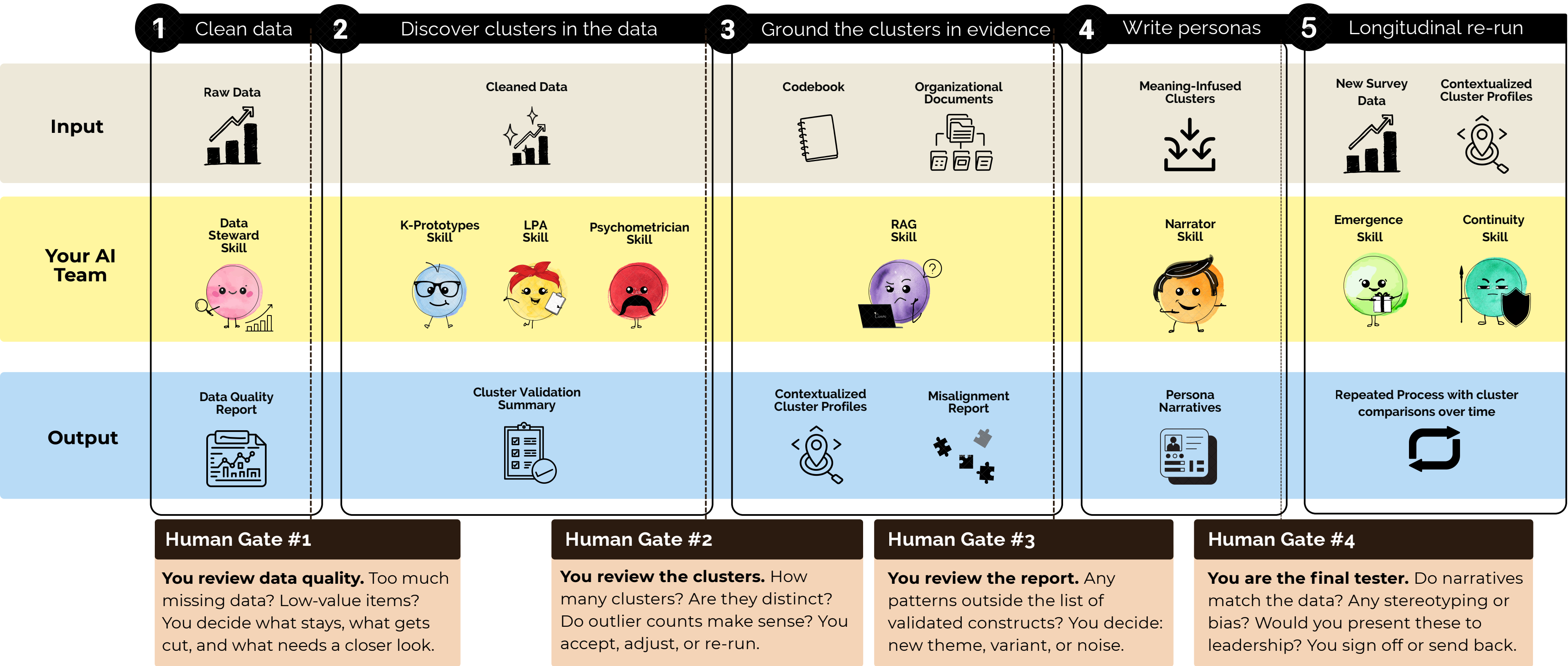
Human in the loop: The system pauses and waits for your input before proceeding. You're an active checkpoint, not an observer.

Examples of bias and where it might enter

- Input: Who responded? Are certain groups underrepresented in the survey data?
- Clustering: do segments map suspiciously onto protected characteristics? Silhouette scores won't flag this; you have to look.
- Narration: AI-generated text can sound neutral while encoding assumptions.
- Context: are the org documents themselves biased? RAG faithfully retrieves what's there; garbage-in-garbage-out applies to organizational artifacts, too!



End to End Process



Scenario

THE CHALLENGE

Your organization just announced a restructuring. Reporting lines shifted and teams consolidated overnight. 10,000 employees responded to the pulse survey, generating thousands of Likert-scale responses, open-ended comments, and demographic/role data pulled from HRIS.

Leadership is asking the same question everyone asks: "Who are our people right now, and what do they need?" The data exist. The people are in there. But thousands of comments can't be hand-coded before Thursday, and a summary score of 3.4 on engagement won't tell anyone what to do next.

Leadership needs to brief the executive team by next week on where to drive intervention efforts. Your job is to answer that question at scale, in a way that a VP can act on.



| PHASE 1

Ingest and Clean Data



| PHASE 1

Ingest and Clean Data

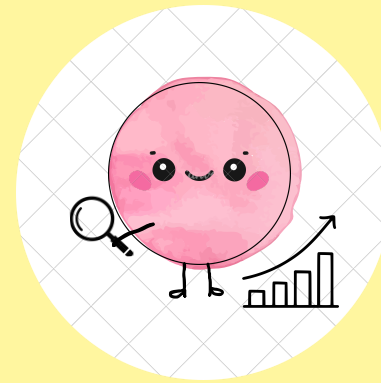
Garbage in, garbage out. This step ensures you don't build personas on noise..

Raw Data



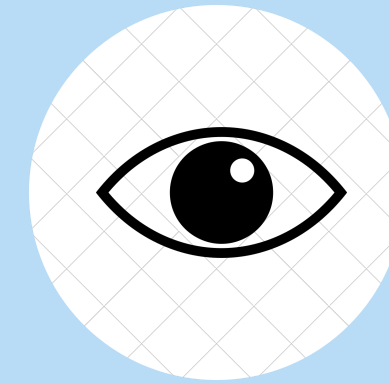
Survey responses
(unstandardized), open-
ended comments, HRIS
demographics

Data Steward Skill

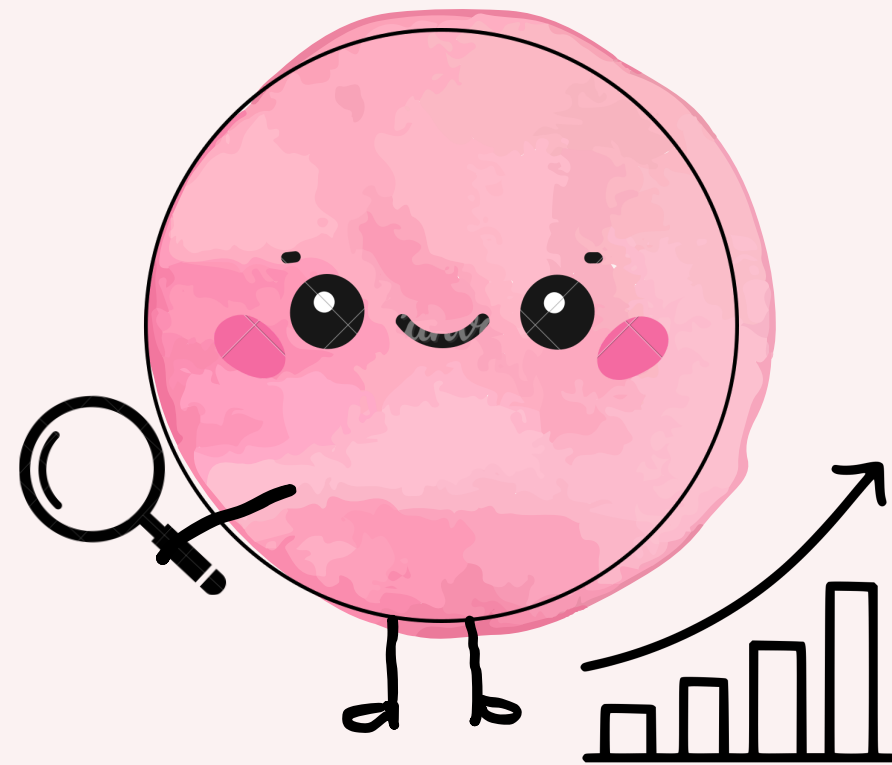


Checks structure.
Removes careless
responders. Flags gaps
>20%. Excludes low-
variance items ($SD < 0.5$).
Flags distribution issues.

Phase 1 Gate



You review data quality.
Too much missing data?
Low-value items? You
decide what stays, what
gets cut, and what needs
a closer look.



Scenario

10,000 employees responded. The Data Steward runs five quality checks. 249 people get flagged for low-quality responses. The bias check shows that removing them wouldn't unfairly affect any group. Data quality confidence comes in at 0.975. You decide to keep all 10,000 to preserve the sample size. Clean data moves to Phase 2.

Phase 1 – GitHub Code

Design decision: The Data Steward does *not* standardize scores. That is deliberately delegated to downstream agents (K-Prototypes and LPA), because each algorithm may need different scaling. This prevents double-standardization errors.

The Data

The synthetic dataset represents **Meridian Technologies**, a mid-size company undergoing a major reorganization. The survey measures five constructs on Likert scales:

Column	Construct	What It Measures
Cared_About	Perceived Organizational Support	"I feel the organization cares about me"
Excited	Work Engagement	"I am excited about the work I do"
Helpful_Info	Communication Effectiveness	"I receive helpful information about changes"
Trust_Leadership	Trust in Leadership	"I trust senior leadership's decisions"
Morale	Work Engagement (affective)	"Overall morale in my team is good"

Plus four demographic columns: Business Unit, Level, FLSA, Tenure. The raw data also includes Attention check 1, Attention check 2, and Comments columns, which the Data Steward uses for careless responding detection before dropping them.

Survey data files:

- synthetic_data/survey_baseline.csv — baseline wave (used for all phases)
- synthetic_data/survey_followup.csv — follow-up wave (Bonus: Longitudinal phase only)

```
# Load the baseline and follow-up surveys
baseline_df = pd.read_csv(os.path.join("../", "synthetic_data", "survey_baseline.csv"))
```

Phase 1 Outputs

Phase 1: Ingest and Clean

Agent: Data Steward
Run ID: fresh_run_2026_04_26
Timestamp: 2026-04-26
Schema Version: 2.0
Mode: Pipeline | INITIALIZATION_MODE = True
Framework: Survey Data Quality Evaluation Model (Papp et al., 2026); Osborne (2013)

Executive Summary

The baseline survey dataset (N=10,000) passed all automated quality gates with a data quality confidence of **0.975** (above the 0.90 pipeline threshold). The dataset is clean and ready for downstream clustering — pending one user decision on careless respondent removal (249 flagged, 2.5%).

All 5 survey items are retained for clustering. No items failed the variance gate. Median imputation was applied to 274 missing cells (<1% of observations). The cleaned dataset contains 10,000 rows with original Likert-scale values (not standardized — standardization is delegated to K-Prototypes and LPA agents).

Key Findings

Gate	Result	Detail
Schema Validation	PASSED	12 columns confirmed; all expected fields present
Careless Responding (SDQEM)	249 flagged (2.5%)	Awaiting removal approval
Sparsity (>20% missing)	PASSED	Max missing: 0.6% (Helpful_Info)
Variance (SD < 0.5)	PASSED	All 5 items retained (SDs: 1.09-1.29)
Distribution Screening	PASSED with notes	Mild IQR artifact on Cared_About, Helpful_Info (Likert scale artifact, not quality concern)
Bias Audit	PASSED	Max demographic shift from removal: 0.1pp
Data Quality Confidence	0.975	Above 0.90 threshold

Phase 1 Outputs

DECISION REQUIRED: Careless Respondent Removal

249 respondents (2.5%) were flagged on >=2 independent SDQEM indicators:

Indicator	Flagged	%
Attention check failures (>=2)	38	0.4%
Straight-lining (longstring >=3 of 5 items)	1,714	17.1%
Low IRV (SD across items < 0.2)	249	2.5%
Multi-hurdle (>=2 indicators)	249	2.5%

The bias audit confirms removal is demographically neutral (max shift: 0.1pp across all groups).

Awaiting Chelsea's decision: Remove the 249 flagged respondents and proceed with N=9,751? Or retain all 10,000?

Survey Item Descriptives (Post-Imputation, N=10,000)

Item	Mean	SD	Median	Notes
Cared_About	3.55	1.15	4.0	Above midpoint
Excited	2.76	1.29	3.0	Below midpoint
Helpful_Info	3.59	1.15	4.0	Above midpoint
Trust_Leadership	2.67	1.29	2.0	Below midpoint — lowest scoring item
Morale	3.18	1.09	3.0	Near midpoint

Interpretation: Trust_Leadership and Excited are the lowest-scoring dimensions, both below the scale midpoint of 3.0. This suggests employees feel reasonably informed (Helpful_Info) and valued (Cared_About) but report lower engagement and limited confidence in leadership. This pattern provides a meaningful backdrop for cluster interpretation in Phase 2.

Artifacts Produced

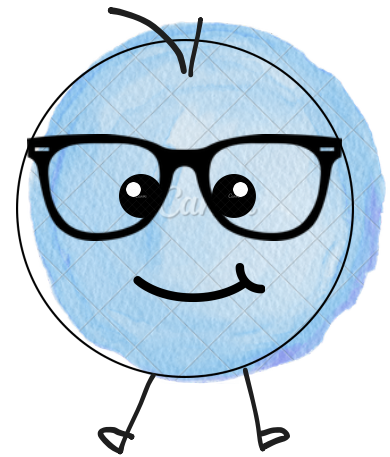
Artifact	Location
Clean data (with careless flag)	outputs/phase1_data_quality_report/survey_baseline_clean.csv
Quality screening report	outputs/phase1_data_quality_report/audit_reports/data_quality_screening.md
Bias audit memo	outputs/phase1_data_quality_report/audit_reports/data_steward_bias_audit.md

| PHASE 2

Discover Workforce Segments



Discover Workforce Segments



K-Prototypes Skill

- Clusters people using both numeric scores AND categorical variables (e.g., department, tenure)
- Assigns each person to exactly one segment based on overall similarity
- Answers: Who are these people demographically and behaviorally?



LPA Skill

- Clusters people using psychometric scores only
- Assigns probabilistic membership - for example, someone can be 70% Profile A, 30% Profile B
- Ambiguity = a finding
- Answers: What psychological profiles exist beneath the surface?

Run both methods

independently to check the validity of your clusters.

- **Agreement** = confidence in your clusters
- **Disagreement** = worth investigating; doesn't necessarily mean clusters are wrong!

| PHASE 2

Discover Workforce Segments

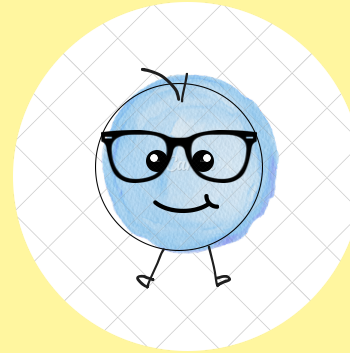
Clustering methods run simultaneously. Convergence builds confidence. Divergence is a finding.

Cleaned Data



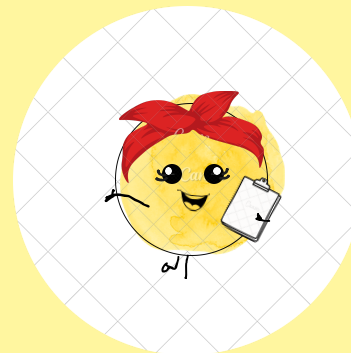
Bad responses removed.
Ready for analysis.

K-Prototypes Skill



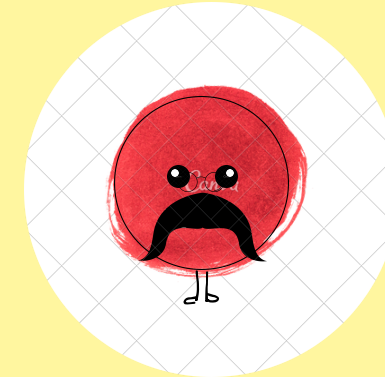
Finds behavioral -
demographic
segments using
distance measures.
**Answers: who are
these people?**

LPA Skill



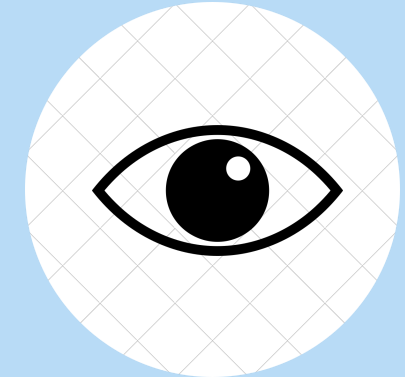
Identifies psychometric
subpopulations via
clusters and assigns
probabilistic
membership.
**Answers: what
psychological profiles
exist?**

Psychometrician Skill

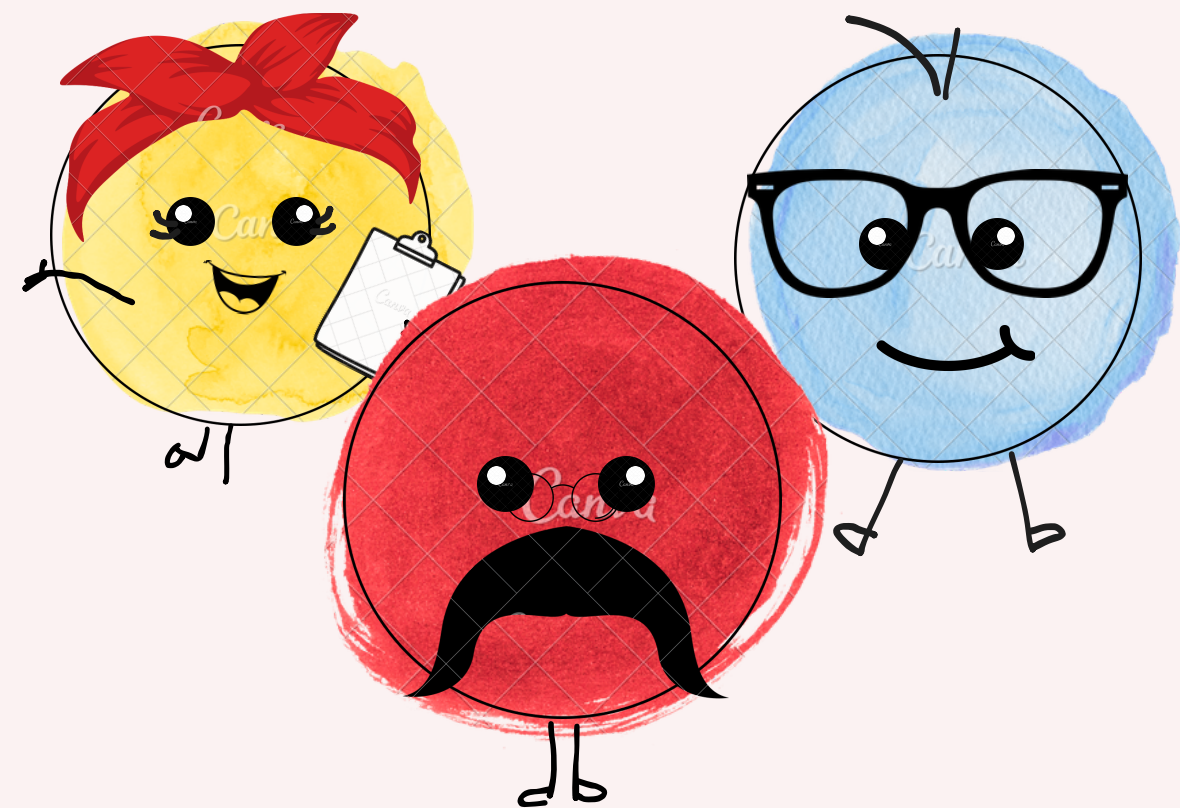


Validates both sets of
groups. Checks if they're
well-separated, flags
outliers, and compares
the two methods.

Phase 2 Gate



You review the findings.
Do K-Proto and LPA agree
on K? What is the cross-
method ARI? What is the
LPA ambiguity rate? Do K-
Proto clusters survive
bootstrap stability? Are
outliers flagged similarly?



Scenario

K-Prototypes and LPA run at the same time, independently. Both methods find the same three groups. Their agreement score is 0.880, which is very high. The two methods match on 93 to 98 percent of people. One technical flag comes up about how distance is calculated on mixed data, but the other validation checks confirm the three groups are real. There's also a hint that one of the three groups might split into two later, worth watching in the next survey. For now, three groups hold. You approve.

Phase 2 – GitHub Code

Key parameters:

- *BIC*: Bayesian Information Criterion -- the primary model selection tool. Balances fit against complexity.
- *Entropy*: Classification certainty. 1.0 = everyone clearly belongs to one group. <0.60 = substantial overlap.
- *Posterior probability*: Each person gets a probability of belonging to each profile. Below 0.70 = "ambiguous."

What it answers: *What psychological profiles exist?* -- segments defined purely by attitude patterns.

The Psychometrician: Cross-Validation

After both methods run, the Psychometrician agent computes:

- **Silhouette scores**: How well-separated are the clusters? (Rousseeuw, 1987)
- **Adjusted Rand Index (ARI)**: How much do the two methods agree? Interpreted through an inter-rater reliability framework (Hallgren, 2012) -- the two algorithms are treated as independent "raters" classifying respondents.
- **Outlier flags**: Respondents far from any cluster centroid who may represent unmodeled experiences.

Interpretation guide for ARI:

- >0.65 = Strong agreement (demographics and psychology tell the same story)
- 0.30-0.65 = Moderate (partially overlapping, which is often the most informative finding)
- <0.30 = Weak (the two methods see very different structure)

```
# --- K-Prototypes: Mixed-type clustering ---
print("Running K-Prototypes (this groups by demographics AND survey scores)...")
kproto_result = run_k_prototypes(clean_df)
audit_trail.extend(kproto_result["audit_entries"])

print(f"\nSelected K = {kproto_result['k_selected']}")
print(f"\nCluster sizes:")
for _, row in kproto_result["profiles"].iterrows():
    print(f"  Cluster {int(row['cluster'])}: n={int(row['n'])} ({row['pct']}%)")
print(f"\n--- Agent Reasoning ---")
```


Phase 2 Outputs

Phase 2: Discover Workforce Segments

Agents: K-Prototypes, LPA, Psychometrician

Run ID: fresh_run_2026_04_26

Timestamp: 2026-04-26

Executive Summary

Two independent clustering methods — K-Prototypes (mixed-data, demographic-aware) and LPA (psychometric-only, model-based) — independently discovered the same three workforce segments at K=3. The Adjusted Rand Index of **0.880** confirms near-perfect agreement between methods. Three validated personas are ready for Phase 3 grounding.

K-Prototypes Results

Cluster	n	%	Fingerprint	Demo Mode
0	3,426	34.3%	Hi-Trust / Hi-Excited / Hi-Morale	Product, Senior, Exempt, 10+ yrs
1	2,949	29.5%	Lo-Caring / Lo-Info / Lo-Morale	Sales, Entry, Non-Exempt, <1 yr
2	3,625	36.2%	Hi-Info+Caring / Lo-Trust+Excited	IT, Mid, Exempt, 3-5 yrs

Bootstrap stability: ARI = 0.986 ± 0.012 (Highly stable). Gamma = 1.0 (Huang default). Cao initialization, n_init=10.

LPA Results

Profile	n	%	Fingerprint	Posterior
0	3,128	31.3%	Lo-Cared_About / Lo-Helpful_Info / Lo-Morale	0.953
1	3,435	34.4%	Hi-Helpful_Info+Caring / Lo-Trust+Excited	0.942
2	3,437	34.4%	Hi-Trust / Hi-Excited / Hi-Morale	0.960

Model: K=3 spherical GMM. BIC=125,399. Entropy=0.881. Ambiguous (post <0.70): 519 (5.2%). K=4 rejected: 4th profile n=76 (0.8%) — outlier junk class per Spurk et al. (2020).

Phase 2 Outputs

Psychometrician Validation

Silhouette Analysis

Metric	Value	Interpretation
Gower silhouette (mixed, subsample N=2000)	0.2013	FAIR — see note
Numeric-only silhouette, K-Proto (Euclidean)	0.3208	FAIR
Numeric-only silhouette, LPA (Euclidean)	0.3229	FAIR

Note on Gower silhouette: The lower value (0.201 vs 0.321) reflects a methodological characteristic of Gower distance on mixed data — categorical mode mismatches inflate distances even for psychometrically well-assigned respondents. The numeric-only silhouette (0.321) is the substantively meaningful metric for psychometric profile separation.

Per-Cluster Scorecard

Cluster	n	Mean Silhouette	Negative Sil %	Outlier %	Grade
0 (Thriving)	3,426	0.311	3.3%	10.9%	B
1 (Disengaged)	2,949	0.248	8.2%	15.5%	C
2 (Informed-Skeptical)	3,625	0.389	1.4%	4.7%	B

Cluster 1 (Disengaged) has the most internal heterogeneity (Grade C, 15.5% outlier rate) — expected for a "catch-all" distressed segment. Cluster 2 (Informed-Skeptical) is the most cohesive (Grade B, silhouette=0.389, only 4.7% outliers).

Cross-Model Validation (ARI)

ARI = 0.880 — STRONG agreement (Hubert & Arabie, 1985)

K-Prototypes	->	LPA Profile	Match Rate
Cluster 0 (Thriving)	→	Profile 2	97.1%
Cluster 1 (Disengaged)	→	Profile 0	98.0%
Cluster 2 (Informed-Skeptical)	→	Profile 1	92.9%

Both independent methods agree on 97-98% of assignments for the Thriving and Disengaged clusters. The Informed-Skeptical cluster shows the most cross-method flux (7.1% disagreement), consistent with it being the most ambiguous boundary between "engaged but frustrated" and "disengaged."

Phase 2 Outputs

Consistency Flag Resolution

A consistency flag was triggered (low Gower silhouette + high ARI). **This is not a quality failure.** The explanation is methodological: Gower silhouette on mixed data systematically undersells psychometric separation because categorical mismatches add Hamming distance even for respondents with ideal psychometric profiles. The numeric silhouette (0.321) and ARI (0.880) together confirm real, replicable structure. No halt warranted; structure is validated.

Outlier Bias Audit

No demographic group exceeds the 1.5x overrepresentation threshold in the outlier pool. Operations (1.32x) and Commercial Department (1.23x) are mildly elevated but well below the concern threshold.

Consolidated Persona Map

Persona	K-Proto	LPA	n (avg)	Psychometric Signature
Thriving	Cluster 0	Profile 2	~3,430 (34%)	High Trust + Excited + Morale
Disengaged	Cluster 1	Profile 0	~3,040 (31%)	Low Caring + Info + Morale
Informed-Skeptical	Cluster 2	Profile 1	~3,530 (35%)	High Info+Caring, Low Trust+Excited

Artifacts

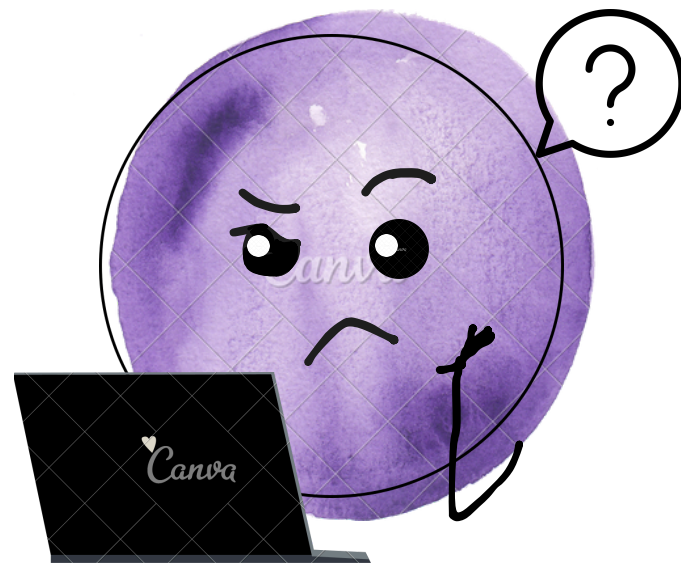
File	Description
cluster_labels.csv	K-Prototypes assignments (N=10,000)
kproto_centroids.json	K=3 numeric + categorical centroids
kproto_profiles.csv	Per-cluster centroid profiles
lpa_profiles.csv	LPA assignments + posterior probabilities
lpa_fingerprints.json	Psychological fingerprints
psychometrician_audit.csv	Per-respondent centroid distance + silhouette
psychometrician_scorecard.csv	Per-cluster validation grades
audit_reports/figures/psychometrician_silhouette.svg	Silhouette plot
audit_reports/figures/psychometrician_crossmodel.svg	ARI heatmap
audit_reports/figures/psychometrician_distances.svg	Distance distribution

| PHASE 3

Ground in Organizational Reality



Ground in Organizational Reality



RAG Skill

The problem without Retrieval Augmented Generation (RAG):

- LLMs generate fluent, confident text
- But without grounding, personas can sound right and be wrong
- Nothing ties the narrative back to your organization's actual context

What RAG does:

- Indexes your org documents (memos, FAQs, policies) into a searchable vector store
- Retrieves relevant passages before generating any text
- Every claim in the persona traces back to a real document

Why it matters for personas:

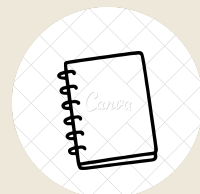
- Prevents hallucination masquerading as insight
- Surfaces gaps between, e.g., what leadership communicated and what employees think and experience
- Makes outputs auditable. You can always ask, "Where did this come from?"

| PHASE 3

Ground in Organizational Reality

Clusters are statistical abstractions. Without organizational context, personas can sound right but be wrong. This phase anchors every claim in real documents.

Codebook*



12 validated I-O constructs with definitions, examples, and non-examples. Constructs are translated into structured queries via the vector store. Governs query construction and isn't embedded from other documents.

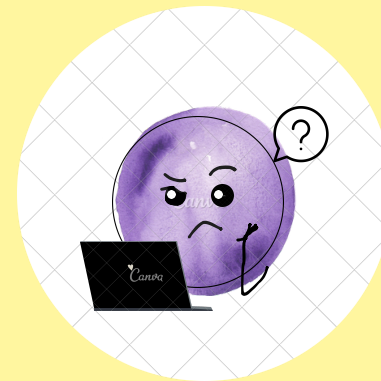
* customizable

Org Docs



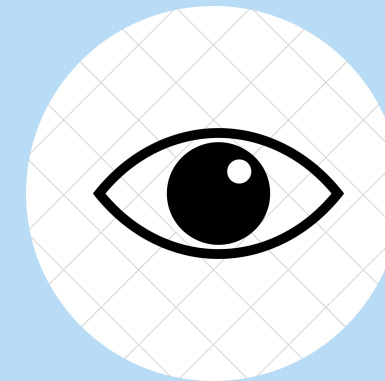
Real organizational documents: restructuring announcements, memos, FAQs, team charters, strategic plans. These are what employees were told. The RAG skill compares them against what employees actually experienced.

RAG Skill



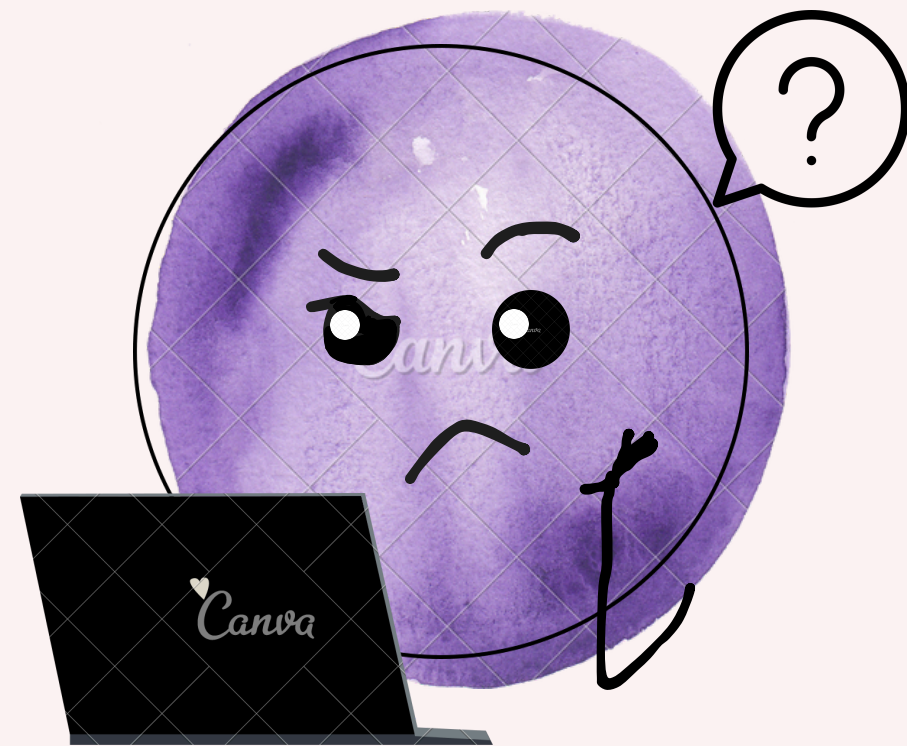
Indexes org documents (policies, memos, FAQs). Uses cluster fingerprints and construct names to retrieve relevant passages. Retrieves actual policy text so personas reference real documents. Flags outdated or contradictory policies. Chunks documents into 500-character windows with 20% overlap

Phase 3 Gate



You review the report.

Any response patterns outside the 12 codebook constructs? You decide: new theme, variant, or noise. Must assess grounding quality as no automatic relevance gate at query time.



Scenario

The Disengaged group scores low on feeling cared for, feeling informed, and morale. The RAG skill pulls the relevant policies: the Reorganization FAQ on mental health support, the team charter's "Put People First" commitment, and the restructuring communication plan. The policies exist. The employees aren't feeling them. That gap, between what leadership said and what employees experienced, is the real story this persona needs to tell. You approve the grounding.

Phase 3 – GitHub Code

Implementation: TF-IDF vectorization with cosine similarity retrieval. This is deliberately simple (avoids PyTorch/embedding model dependencies). A production system would use dense embeddings for better semantic matching, but TF-IDF is sufficient for this tutorial and keeps the setup lightweight.

For background on RAG, LLMs, and embeddings, see [resources/ai_foundations/key_concepts.md](#).

Emergence Agent (Cross-Sectional Mode)

The I-O codebook defines 12 validated constructs. But what if the data contains meaningful patterns that fall *outside* those 12? The Emergence Agent scans cluster profiles for unusual construct combinations and classifies each candidate as:

- **NEW:** A genuinely novel theme not captured by the codebook
- **VARIANT:** A known construct appearing in an unusual combination
- **NOISE:** A statistical artifact, not a meaningful pattern

This operationalizes the grounded theory principle (Glaser & Strauss, 2017) that the codebook catches what we *expect* to find, and emergence detection surfaces what we did *not* expect.

```
# Build the knowledge base from organizational documents
print("Building knowledge base from organizational documents...")
kb = build_knowledge_base(os.path.join("../", "synthetic_data", "org_documents"))
print(f"Loaded {kb['n_documents']} documents, created {kb['n_chunks']} searchable chunks.")

# Demonstrate retrieval
print("\n--- Retrieval Demonstration ---")
print("Querying the knowledge base to show what RAG finds:\n")

demo_queries = [
    "employee trust in leadership during reorganization",
    "remote work policy and flexibility",
    "career development and growth opportunities",
]
for q in demo_queries:
    results = query_knowledge_base(kb, q, top_k=2)
```


Phase 3 Outputs

6 documents indexed | 72 chunks | 384-dimension embeddings | 0 coverage gaps

All 8 expected policy topic areas covered. No outdated documents flagged. No contradiction candidates identified.

Per-Cluster Policy Grounding

Cluster 0 — Disengaged (Low Caring / Low Info / Low Morale)

Top retrieved policy snippets:

- `employee_faq.md` [Reorganization FAQ] — Employee Assistance Program and mental health resources (similarity 0.353)
- `team_charter.md` [Put People First] — Human impact alongside business impact commitment (0.329)
- `team_charter.md` [Our Commitment During This Transition] — Feedback channels and accountability pledges (0.348)
- `restructuring_announcement.md` [Timeline] — Transition milestones and communication schedule (0.316)

Policy context: The organization explicitly acknowledges employee stress during restructuring and has committed to open feedback channels. The gap between these stated commitments and the Disengaged cluster's experience (low trust in leadership, low sense of being cared for) is the central tension this persona surfaces.

Phase 3 Outputs

Cluster 1 — Informed-Skeptical (High Info+Caring / Low Trust+Excited)

Top retrieved policy snippets:

- `team_charter.md` [Build Trust Through Transparency] — Core values on earning trust through transparency (0.480)
- `restructuring_announcement.md` [Timeline] — Specific milestones and integration schedule (0.603)
- `strategic_direction.md` [Growth Targets] — Concrete metrics and internal mobility goals (0.603)
- `strategic_direction.md` [How You Fit In] — Individual role clarity within reorganization (0.580)
- `employee_faq.md` [Reorganization FAQ] — Job security and career path continuity (0.419)

Policy context: This cluster scores highest on relevance to org documents (multiple hits >0.50). These employees are well-informed about the restructuring and strategic direction — which is exactly what the data show (High Helpful_Info). The skepticism (Low Trust, Low Excited) coexists with genuine organizational awareness. Leadership follow-through on stated transparency commitments is the key lever for this group.

Cluster 2 — Thriving (High Trust / High Excited / High Morale)

Top retrieved policy snippets:

- `strategic_direction.md` [Investment Areas] — Engineer training programs and leadership development (0.470)
- `team_charter.md` [Grow Together] — Learning as part of the job; individual and org growth inseparable (0.513)
- `strategic_direction.md` [Investment Areas] — Customer retention strategy and employee empowerment (0.467)
- `team_charter.md` [Put People First] — Values alignment and human impact (0.454)
- `team_charter.md` [Our Commitment] — Transition commitments maintained/expanded (0.419)

| PHASE 4

Write and Validate the Persona

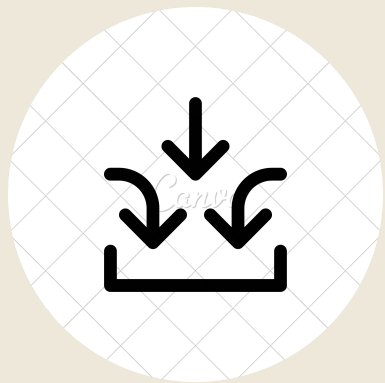


| PHASE 4

Write and Validate the Persona

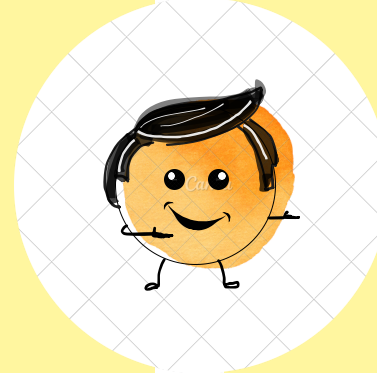
This is where data becomes something leaders can act on. Every persona narrative is constrained by the codebook and grounded in evidence. No claim without a source.

Narrator Inputs



Cluster labels. Profiles.
Policy context. Codebook.
Comments.

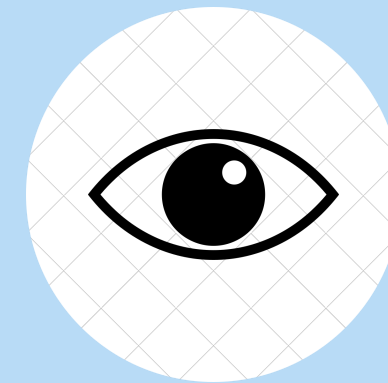
Narrator Skill



Writes a persona for each segment. Combines who they are, how they think, and what the org. says. Flags mismatches between policy content and cluster fingerprints.

Statistical evidence constrains what the narrative can claim. Org. policy grounds evidence via RAG retrieved policies. Individual evidence anchors with quotes.

Phase 4 Gate



You are the final tester.
Do narratives match the data? Any stereotyping or bias? Grounded in thematic analysis principles? Every narrative claim is anchored to respondent data or statistical evidence?



Scenario

The Narrator writes the persona for the Unsupported Strivers. Entry-level Sales employees, less than a year on the job, who feel overlooked. This isn't anger. It's neglect. They don't believe the organization sees them or invests in them, and they don't feel informed about where the company is headed. Their own words: "Pay has not kept up. I feel replaceable." "I don't know what the company's plan is. I'm not sure anyone does." "I don't feel valued. My manager is overloaded and I rarely get feedback." The signal is flight risk. They need early intervention. Quality grade is C, so the persona comes with a caution label. Every claim ties back to the data or a real quote. You review. You note the caveat. You approve.

Phase 4 – GitHub Code

trust, not "feeling betrayed" or "resentful."

2. **Fabricated quotations:** No invented quotes. Every claim must trace to centroid values, fingerprints, or RAG-retrieved passages.
3. **The Oracle Effect:** The narrative must state its uncertainty explicitly rather than masking it with confident language.

Each persona includes:

- A neutral, non-stigmatizing name
- A statistical fingerprint (the numbers behind the narrative)
- Policy-experience mismatch flags (where stated policy contradicts the cluster's lived experience)
- An epistemic note disclosing evidence basis and limitations

For background on how LLMs generate text and why these guardrails matter, see [resources/ai_foundations/ai_models_explained.md](#).

```
# Generate persona narratives
print("Generating persona narratives...")
narrator_result = generate_personas(
    cluster_profiles=kproto_result["profiles"],
    construct_scores=lpa_result.get("fingerprints"),
    policy_context=grounding_result.get("mappings"),
    codebook=list(config.NUMERIC_COLS),
)
audit_trail.extend(narrator_result["audit_entries"])
personas = narrator_result["personas"]

print(f"\n--- Agent Reasoning ---")
print(narrator_result["reasoning"])
```

```
# Display the personas
display(Markdown("## Draft Persona Narratives\n"))

if isinstance(personas, list):
    for i, persona in enumerate(personas):
        if isinstance(persona, dict):
            name = persona.get("persona_name", persona.get("name", f"Persona {i+1}"))
            display(Markdown(f"---\n### {name}"))
```

Phase 4 Outputs

Persona Synthesis Overview — Meridian Technologies

Run ID: fresh_run_2026_04_26 **Timestamp:** 2026-04-26 **Total respondents:** 10,000 **Clusters:** 3 **Global Silhouette (numeric):** 0.3208 (FAIR) **ARI (K-Prototypes vs LPA):** 0.880 (STRONG)

Cluster Comparison

Cluster	Name	Size	%	Grade	Silhouette	Key Signature
0	The Anchored Advocates	3,426	34.3%	B	0.311	Hi-Trust / Hi-Excited / Hi-Morale
1	The Unsupported Strivers	2,949	29.5%	C	0.248	Lo-Caring / Lo-Info / Lo-Morale
2	The Clear-Eyed Skeptics	3,625	36.2%	B	0.389	Hi-Info+Caring / Lo-Trust+Excited

Cross-Model Validation

Both K-Prototypes (mixed-data) and LPA (psychometric-only) independently discovered the same three segments:

K-Prototypes	LPA Profile	Agreement
Cluster 0 (Anchored Advocates)	Profile 2	97.1%
Cluster 1 (Unsupported Strivers)	Profile 0	98.0%
Cluster 2 (Clear-Eyed Skeptics)	Profile 1	92.9%

ARI = 0.880 confirms near-perfect convergence between methods.

Construct Mapping (IO Codebook)

Cluster	Primary Constructs	Disruption Signal
Anchored Advocates	TRUST-LDR (high), WRK-ENG (high), ORG-COM (high)	Resilient; maintain during change
Unsupported Strivers	POS (low), COMM-EFF (low), ORG-COM (low)	Flight risk; need early intervention
Clear-Eyed Skeptics	COMM-EFF (high), JUST-PRO (low), CHG-RDY (low)	Watch-and-wait; trust is the lever

Phase 4 Outputs

Emergence Note

The Emergence Agent classified K=4 as AMBIGUOUS (evidence score = 0.67). The Disengaged/Unsupported Strivers cluster shows internal heterogeneity (Cohen's d = 0.98 between K=4 sub-groups, 100% bootstrap stability). Recommend monitoring this cluster for potential fracture in the next survey wave.

Epistemic Caveats

- 1. All five survey items are single-item proxies for validated multi-item constructs (see IO Codebook). Findings should be interpreted as directional indicators, not precise construct measurements.
- 2. The Unsupported Strivers cluster (Grade C) has the weakest statistical support and highest internal variability. Characterizations of this cluster require particular caution.
- 3. Demographic tendencies (modal Business Unit, Level, etc.) describe probabilistic patterns, not deterministic group membership. Any given employee may belong to any cluster.
- 4. The IO Psychologist retains final interpretive authority over all persona characterizations.

Generated with AI assistance. All narrative claims traceable to statistical centroid values. Quotes are verbatim; no fabrication, paraphrase, or synthesis (Nguyen & Welch, 2025).

Phase 4 Outputs

Cluster 1: The Unsupported Strivers

Run ID: fresh_run_2026_04_26 | Timestamp: 2026-04-26

Statistical Fingerprint

Dimension	Z-score	Interpretation
Cared_About	-1.063	Low
Excited	-0.337	Moderate
Helpful_Info	-1.039	Low
Trust_Leadership	-0.347	Moderate
Morale	-0.883	Low

Size: 2,949 respondents (29.5% of total) **Quality Grade:** C [interpret with caution] (Silhouette = 0.248) **Outlier rate:** 15.5% **Dominant demographic:** Business Unit: Sales | Level: Entry | FLSA: Non-Exempt | Tenure: < 1 yr

LPA Psychological Fingerprint: Low-Cared_About / Low-Helpful_Info / Low-Morale

Narrative

This cluster captures employees who feel underserved by the organization’s support and communication systems. They score below the sample mean on Cared_About (-1.06 SD), Helpful_Info (-1.04 SD), and Morale (-0.88 SD), while Excited (-0.34 SD) and Trust_Leadership (-0.35 SD) fall in the moderate-low range. The pattern is not one of active hostility but of perceived neglect: these respondents lack confidence that the organization sees or invests in them, and they report limited access to information about organizational direction. Their demographic profile (Entry-level, Sales, Non-Exempt, under 1 year tenure) flags a potential gap in onboarding and early-career support. This cluster carries the highest internal heterogeneity (Grade C, silhouette=0.248) and highest outlier rate (15.5%), indicating that disengagement in this segment is not a single phenomenon but a family of related experiences. Interpret this cluster with appropriate caution given its Grade C quality rating.

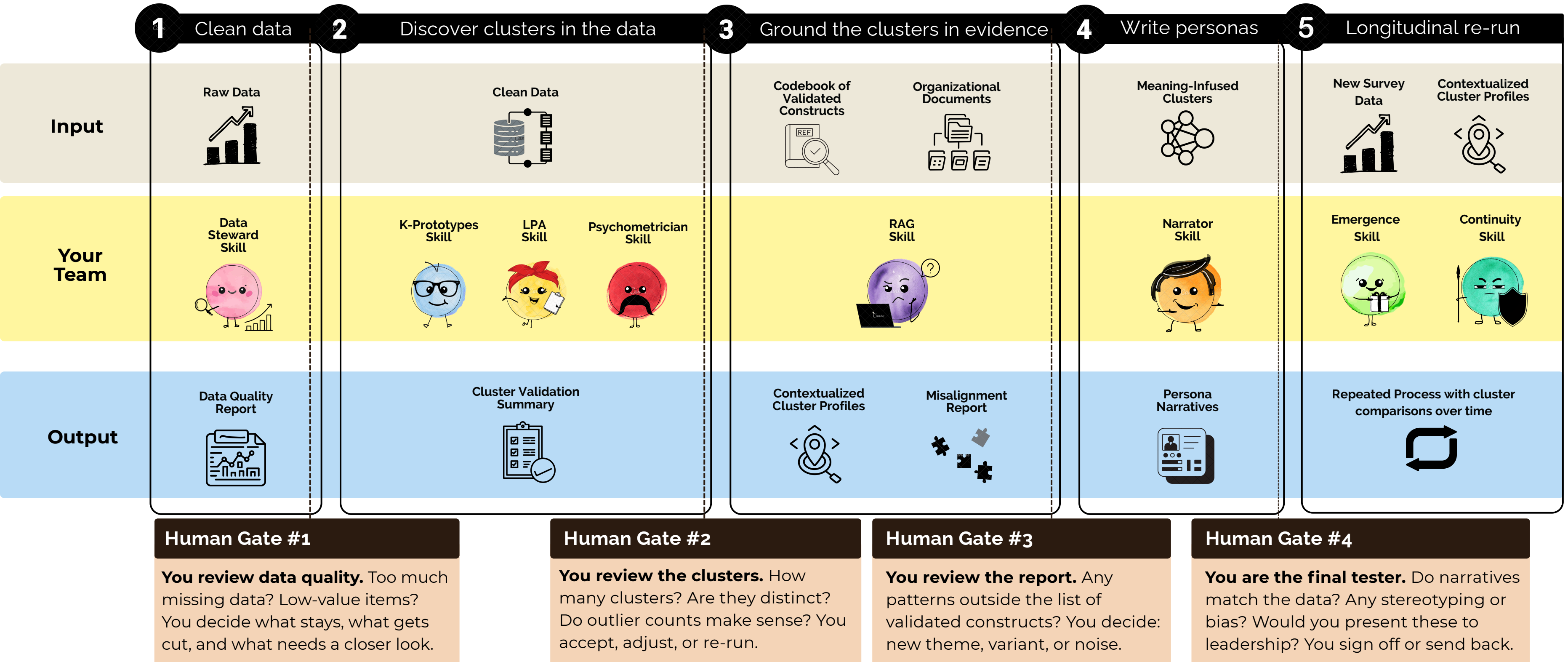
Representative Voices

"Pay has not kept up. I feel replaceable." *(Verbatim respondent comment; non-outlier; centroid distance = 0.579)*

"I don't know what the company's plan is. I'm not sure anyone does." *(Verbatim respondent comment; non-outlier; centroid distance = 0.579)*

"I don't feel valued. My manager is overloaded and I rarely get feedback." *(Verbatim respondent comment; non-outlier; centroid distance = 0.579)*

End to End Process



Cluster 0: The Anchored Advocates

Run ID: fresh_run_2026_04_26 | Timestamp: 2026-04-26

Statistical Fingerprint

Dimension	Z-score	Interpretation
Cared_About	+0.356	Moderate
Excited	+1.026	High
Helpful_Info	+0.170	Moderate
Trust_Leadership	+1.031	High
Morale	+0.741	High

Size: 3,426 respondents (34.3% of total) **Quality Grade:** B (Silhouette = 0.311) **Outlier rate:** 10.9% **Dominant demographic:** Business Unit: Product | Level: Senior | FLSA: Exempt | Tenure: 10+ yrs

LPA Psychological Fingerprint: High-Trust_Leadership / High-Excited / High-Morale

Thank you!

QUESTIONS?

